Evaluation of Measurement Uncertainty

[In accordance with ISO Guide 98-3: 2008]

DETERMINATION OF FLOW OF GROUT FOR PREPLACED - AGGREGATE CONCRETE (FLOW CONE METHOD)
ASTM C939:2022

1 **Test equipment**

Stop Watch

Flow cone

2 **Mathematical model**

Time of flux (T) = rmeasured time (MR)

3 **Uncertainty equation**

T = MR

u_c^2(T) = [c(C) · u(C)]^2 + [c(Res) · u(Res)]^2+ [c(Rep) × u(Rep)]^2

with the sensitivity coefficient:

c(C), c(Res) and c(Rep) =

4 **Source of uncertainty**

where C = calibration

Res = resolution

Rep = repeatability

5 **Input data**

Stop watch Code:

Calibration uncertainty	(s)		(k =		)
Resolution	(s)				
Error	(s)				
Repeatability	(s)	1.8			

6 Evaluation of measurement uncertainty (T)

Factor	a	Distribution	ci (sensitivity coeff.)	u(xi)*ci	ci*u(xi)^2	γ	ci*u(xi)^4/γ
Calibration	#DIV/0!	normal	1	#DIV/0!	#DIV/0!	10000	#DIV/0!
Resolution	0	rectangular	1	0.000000	0.00000	10000	0.000E+00
Error	0	rectangular	1	0.000000	0.00000	10000	0.000E+00
Repeatability	1.0392305	rectangular	1	1.039230	1.08000	10000	1.166E-04
				sum	#DIV/0!		#DIV/0!

uc	#DIV/0!	
Veff	#DIV/0!	
t value	2	
U95	#DIV/0!	sec